Power Budget Dual Motor Subsystem

Team Number:	304
Project Name:	EV-Scope (Motors)
Team Member Names:	Quinn, Roshan, Dylan
Version:	2

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,

All Major Components	Component Name	Part Number	Supply	#	laximumCurr	Total	Unit
	3.3-5V Regulator	LM2575	+4.75 - 40V	1	800	800	mA
	Stepper motor	324	+12 - 24V	2	350	700	mA
	Motor Driver	TMC2209	+4.8 - 5.25V	2	10	20	mA
	Microcontroller	PIC18F57Q43	+1.8 - 5.5V	1	350	350	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails

+12V Power Rail	Component Name	Part Number	lyVoltageRa	#	laximumCurr	talCurrent(mA	Unit
	Stepper motor	324	+12 - 24V	2	350	700	mA
					Subtotal	700	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	875	mA
c1. Regulator or Source Ch	Barrel Jack	10811	+9 - 12V	1	4000	4000	mA
					Total Remaining Current Available on +12V Rail	3125	mA
+5.0V Power Rail	Component Name	Part Number	lyVoltageRa	#	laximumCurr	talCurrent(mA	Unit
	Motor Driver	TMC2209	+4.8 - 5.25V	2	10	20	mA
	Microcontroller	PIC18F57Q43	+1.8 - 5.5V	1	350	350	mA
						0	mA
						0	mA
						0	mA
					Subtotal	370	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	462.5	mA
c2. Regulator or Source Ch	+5.0V Regulator	LM2575	+4.75 - 40V	1	1000	1000	mA
					Total Remaining Current Available on +3.3V Rail	537.5	mA

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all

External Power Source 1	Component Name	Part Number	lyVoltageRa	Outpu	laximumCurr	talCurrent(mA	Unit
Power Source 1 Selection	Barrel Jack	10811	+9 - 12V	1	4000	5000	mA
	+5.0V Regulator	LM2575	+4.75 - 40V	1	1000	1000	mA
					Total Remaining Current Available on External Power Source 1	4000	mA